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AGGN: Attention-based glioma grading network with multi-scale feature extraction and multi-modal information fusion☆

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ABSTRACT

In this paper, a magnetic resonance imaging (MRI) oriented novel attention-based glioma grading network (AGGN) is proposed. By applying the dual-domain attention mechanism, both channel and spatial information can be considered to assign weights, which benefits highlighting the key modalities and locations in the feature maps. Multi-branch convolution and pooling operations are applied in a multi-scale feature extraction module to separately obtain shallow and deep features on each modality, and a multi-modal information fusion module is adopted to sufficiently merge low-level detailed and high-level semantic features, which promotes the synergistic interaction among different modality information. The proposed AGGN is comprehensively evaluated through extensive experiments, and the results have demonstrated the effectiveness and superiority of the proposed AGGN in comparison to other advanced models, which also presents high generalization ability and strong robustness. In addition, even without the manually labeled tumor masks, AGGN can present considerable performance as other state-of-the-art algorithms, which alleviates the excessive reliance on supervised information in the end-to-end learning paradigm.

1. Introduction

As one of the most common primary tumors caused by the cancerization of glial cells in the brain or spinal cord, glioma accounts for nearly half of intracranial tumors and 36% of the nervous system tumors [1]. According to the criteria of the World Health Organization (WHO), glioma can be graded as four levels from I to IV [2], where low-grade glioma (LGG) includes grades I–II, and grades III–IV are the so-called high-grade glioma (HGG). It is worth mentioning that LGG may be cured by drug therapy and surgical excision, whereas radiotherapy and chemotherapy are required to cure HGG. Moreover, patients with the glioma of grade IV even suffer from low survival rate less than 10% [3]. Consequently, it is of vital significance to realize accurate preoperative grading of glioma.

In clinical practice, the glioma grading task is mainly accomplished by imaging diagnosis [4–6], including computed tomography (CT) [7], magnetic resonance imaging (MRI) [8] and positron emission tomography (PET) [9], etc. Particularly, the MRI has advantages of strong specificity and sensitivity in tumor localization and pathological analysis [10], which can generate multi-modal images to reflect brain feature information at different levels by modulating imaging parameters. To be specific, four modalities are included in MRI [11], where the T1-weighted (T1) modality displays brain anatomy, and the T2-weighted (T2) one locates lesion area; fluid-attenuated inversion recovery (FLAIR) and T1-weighted contrast-enhanced (T1ce) modalities are generally used for visualization of peritumor and internal conditions so as to make further pathological analysis [12]. Due to above four modalities can sufficiently present structural and functional information of tumors, the multi-modal MRI technique has become an important diagnostic tool of grading glioma in clinic.

It is worth pointing out that inspecting the diverse information provided by multi-modal MRI is a laborious task, which inevitably increases the workload on radiologists or neurosurgeons. For a clear view, the MRI slices of LGG and HGG are illustrated in Figs. 1(a) and 1(b), respectively. Experienced doctors generally distinguish LGG from HGG by observing the clarity of tumor contour and the presence of

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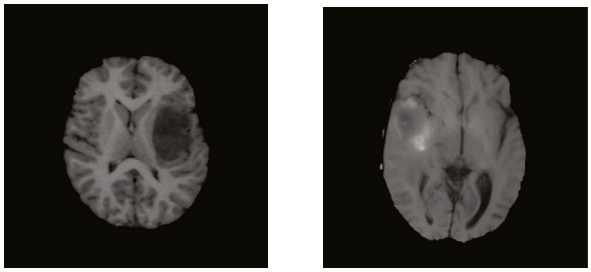


(a) LGG (b) HGG

Fig. 1. MRI visualization of gliomas with different grades.

edema in peritumor areas. Hence, it can be inferred that both the high heterogeneity of brain tumors and experiences of doctors will influence the final diagnostic results. As a result, it is necessary and beneficial to develop computer-aided diagnosis (CAD) systems to realize accurate glioma grading with less manpower [13–15].

Owing to the continuous development of the deep learning (DL) techniques [16,17], plenty of CAD methods have been proposed and applied to the intelligent analysis of brain MRI [18–21], and related studies regarding to glioma grading tasks are reviewed in Section 2. It is noticeable that in the context of applying DL-based models for medical image analysis [22,23]. A common and challenging issue is the robust feature extraction, which has great impact on the downstream tasks (e.g., segmentation and classification). Regarding to the glioma grading task, the above problem is reflected on identifying different modalities with highly similar imaging features, and it is also tough to effectively utilize both semantic and detailed information under different MRI modalities. In addition, most existing glioma grading algorithms have great reliance on the data with manually labeled tumor masks, while it is a time-consuming and laborious task to obtain those masks.

Based on above discussions, in this paper, a novel attention-based glioma grading network (AGGN) is proposed to overcome the mentioned challenges. In particular, by meticulously designing three modules to realize the function of dual-domain attention, multi-scale feature extraction and multi-modal information fusion, the proposed AGGN can efficiently capture potential correlations and key information from scattered features in different imaging modalities, and enhance the ability to distinguish intra-class variability and inter-class similarity existing at different grades of glioma. Based on the final fused highly discriminative features, the proposed AGGN can present comparable grading performance even without the manual labeled tumor masks. In addition, the ability of extracting features with strong presentation also guarantees the model robustness and generalization performance to some extents. Therefore, it is also feasible and promising to apply the developed AGGN into other MRI-based tasks, such as the diagnosis of Parkinson's disease and Alzheimer's disease [24,25]. Major contributions of this paper are outlined as follows:

(1) A novel brain MRI analysis method AGGN is proposed for grading glioma, which can reduce the reliance on supervised information of manual labels.
(2) Designed multi-scale feature extraction and multi-modal information fusion modules benefit extracting discriminative features with strong presentation.
(3) Evaluations on both internal and external brain MRI datasets have demonstrated superiority of the proposed AGGN, which yields satisfactory robustness and generalization ability.

The remainder of this paper is organized as follows. Related work on glioma grading is presented in Section 2. The proposed AGGN and the key components are elaborated in Section 3. Substantial experimental validations and comprehensive discussions are presented in Section 4. Finally, conclusions with an outlook of future work are drawn in Section 5.

2. Related work

In this section, related glioma grading methods are reviewed. As glioma grading is essentially an image classification task, representative feature extraction and fusion methods are briefly introduced as well.

2.1. Glioma grading methods

In clinic, CAD methods play an important role in grading glioma with brain MRI data, and as early as 2010, the authors in [26] have used a support vector machine (SVM) to realize the preliminary assessment of glioma grade and achieved accuracy of 82%. In [27], a two-level clustering method has been proposed for MRI preprocessing, after which an SVM is adopted to accomplish the glioma grading task. A combination of SVM and multi-layer perceptron has been adopted to glioma grading in [28], where feature selection is performed on tumor sub-regions of different modalities. In addition to SVM, other classic machine learning models have also been applied in this area, and one can refer to [29] for more information.

Owing to the rapid development of DL techniques, deep neural networks (DNNs) based glioma grading models have already become the mainstream, where convolutional neural network (CNN) is one of the most popular architectures, including 2D- and 3D-CNN according to dimension of the convolution operations. In [30], a lightweight 2D-CNN model has been developed with only basic components like convolution and pooling layers, and the proposed method has realized fast inference with low computational complexity. In [31], the authors have proposed a 3D-CNN, where the volume of interests is segmented to promote an efficient feature extraction. In particular, performance between 2D Mask R-CNN and 3D U-Net in glioma grading task has been compared in [32], and it is found that 2D model achieves higher sensitivity but lower specificity than the 3D one. Transfer learning paradigm has been introduced in [33], with two well-known CNN-based models AlexNet and GoogleNet, experimental results indicate that the pre-trained model can enhance the performance. Similarly, it is deemed in [34] that the pre-trained CNN model can extract high-dimensional information of feature maps, which benefits further grading of glioma with stronger presentation than the low-dimensional texture or shape features.

In addition, a three-stage DNN model has been developed in [35], which successively performs the rough contour segmentation, the precise contour extraction and the classification. In [36], the generative adversarial network has been utilized to solve the problems of limited samples in brain MRI. Meanwhile, an adaptive encoder has been employed to extract multi-modal features in [36], which finally achieves precision of 92% on the glioma grading task.

Although above methods have proven effective, following two important issues still deserve further improvement. Firstly, most grading models rely on tumor mask-based data, and it is difficult to achieve end-to-end training without manual annotation; secondly, it is of vital significance to efficiently capture and integrate multi-modal pathological glioma features from MRI data, which has not been well addressed in existing methods.

To overcome the mentioned problems, in the proposed AGGN, three modules are meticulously designed to realize dual-domain attention mechanism, multi-scale feature extraction and multi-modal information fusion, so that highly discriminative features with strong presentation can be extracted. Details of the proposed AGGN are presented in Section 3.

2.2. Feature extraction and fusion methods

In the context of DL-based medical image processing, extracted robust features with strong presentation will have a great impact on the model performance. Representative CNN-based feature extractors include the visual geometry group (VGG) [37], GoogleNet [38], Inception

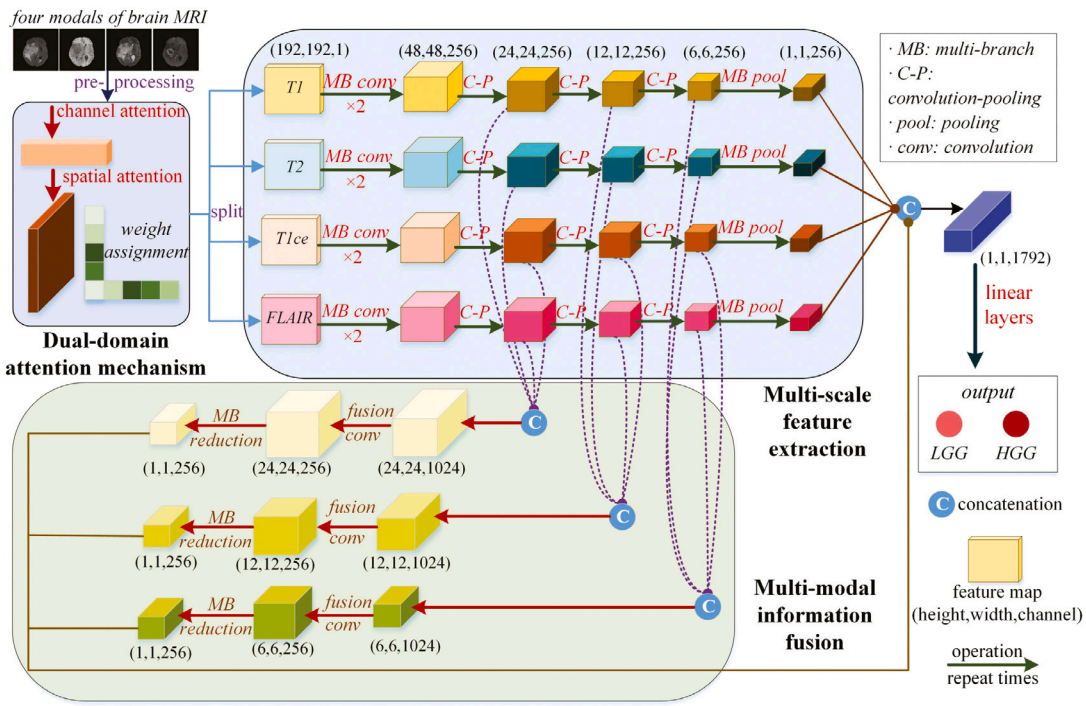


Fig. 2. Framework of the proposed attention-based glioma grading network (AGGN).

v1–v4 [39,40], residual network (ResNet) [41], re-parameterization VGG (RepVGG) [42], etc. In particular, VGG has reduced the amount of model parameters by stacking small convolution modules; GoogleNet and Inception v1 have utilized multi-branch architectures with multi-size convolutions to extract features; Inception v2–v4 models have further proposed the concept of BatchNorm, asymmetric decomposition convolution kernel and residual inception to enhance the performance; ResNet has solved the gradient explosion problem by skip connections in deep structures; in RepVGG, the dominance of both detection speed and accuracy have been achieved by decoupling the training process and inference stage.

Feature fusion is another important operation in many DL-based methods, which promotes sufficient integration of information at different levels so as to enhance the presentation ability of features and improve the model performance. One of the most representative feature fusion structures is the feature pyramid network (FPN) [43], which contains two pathways for bottom-up forward propagation and top-down sampling recovery, respectively, and the lateral connections in FPN have facilitated information fusion. It is noticeable that many FPN variants have been successfully proposed, such as path aggregation network (PANet) [44], bidirectional FPN (BiFPN) [45] and atrous spatial pyramid pooling-balanced FPN (ABFPN) [46], etc. Particularly, the ABFPN is an enhanced multi-scale feature fusion structure, which improves the model performance via sufficiently utilizing context information and generating balanced enhanced features with rich receptive fields.

3. Methodology

In this section, the proposed AGGN is elaborated with implementation details, including the designed dual-domain attention mechanism, multi-scale feature extraction and multi-modal information fusion modules. To begin with, the overall framework of AGGN is illustrated in Fig. 2.

3.1. Overall framework of AGGN

According to Fig. 2, in the proposed AGGN, firstly pre-processing operations including standardization, center cropping, modal splicing, data partitioning and augmentation are performed on the input brain MRI images with four modalities T1, T2, T1ce and FLAIR. Afterwards, the pre-processed multi-modal MRI data will enter the dual-domain attention mechanism module, where the weights are assigned in both channel and spatial dimensions to highlight the key information and suppress the unimportant one in feature maps. Next, the multi-modal MRI is split, and in the followed multi-scale feature extraction module, parallel processing is performed on each single-modality map, including sequential operations of multi-branch convolution (MB conv), convolution-pooling (C-P) and multi-branch pooling (MB pool), and the final output of each pathway is in the size of $1 \times 1 \times 256$. It is noticeable that during above procedure, maps with three sizes on each modality are individually concatenated and fed into the multi-modal information fusion module, which contains the fusion convolution and MB reduction operations. At last, seven feature maps in the same size are cascaded and fed into the linear layers, which is responsible to accomplish the glioma grading task.

In the following subsections, above mentioned three major modules of AGGN are presented with details.

3.2. Dual-domain attention mechanism

Attention mechanism is essentially a procedure of weighting features by pixel-wise operations in channel or spatial dimension, where the position that can reflect the detailed or semantic information of targets will be assigned large weights. In the proposed AGGN, a novel dual-domain attention mechanism is designed, and the structure is presented in Fig. 3. It is worth mentioning that the pre-processed input data are directly sent into the designed dual-domain attention module to model the target location and individual modalities, where different weights are assigned based on both channel and spatial importance of the features, so as to realize focused attention on the useful information and simultaneously suppress the useless one.

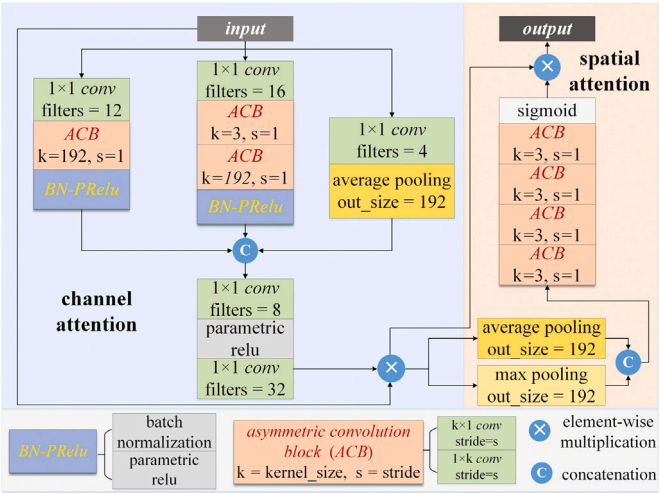


Fig. 3. Dual-domain attention mechanism module.

As is shown in Fig. 3, the input map will successively pass through the channel attention (CA) and spatial attention (SA) components. To be specific, in the former one, the size of feature maps in four modalities is compressed through operations of three parallel branches, where 1×1 convolution, asymmetric convolution block (ACB), average pooling and BN-PRelu (batch normalization and parametric Relu) operations are performed. It is noticeable that ACB replaces square convolution with asymmetric one equivalently [47], which can effectively avoid significant information loss and reduce the number of parameters. By concatenating the three branches, diverse information is shared and afterwards weights are assigned via activation operation and element-wise multiplication with the original input data.

In the subsequent spatial attention component, average and maximum pooling layers are placed at first to compress the channel of feature maps, and the outputs are concatenated to enter series of ACB blocks to learn the parameters in spatial dimension. Similarly, after the sigmoid activation function, the spatial-domain weight assignment for different pixel regions is eventually achieved by element-wise multiplication. As a result, the applied dual-domain attention mechanism can figure out both "where" and "what" the model should pay attention to. Work principle of above module is described by following equation.

$$output_{am} = SA\left\{M\left[CA\left(F_i^c\right)\right]_j^{x,y}\right\}, \quad (i = 1, \ldots, n;\ j = 1, 2, 3, 4) \tag{1}$$

where $F_\cdot^c$ and $M_\cdot^{(x,y)}$ refer to the feature map of channel c and modality of brain MRI in position (x, y), respectively; n is the number of feature maps, j denotes the modal and $output_{am}$ is the final output.

3.3. Multi-scale feature extraction

As previously mentioned, the four MRI modalities have contained rich pathological information with different concerns. To realize sufficient feature extraction on each modality, output of the dual-domain attention module is further split into four single-modal maps to enter the multi-scale feature extraction module (see Fig. 2), which can promote in-depth analysis of the key features enhanced by attention mechanism and can also benefit the subsequent multi-modal information fusion as well. On each branch, the involved MB conv, C-P and MB pool are displayed in Figs. 4(a)–4(c), respectively.

The MB conv block is used to extract the shallow features of each modal. As can be seen from Fig. 4(a), three parallel branches with different operations are included so that the extracted feature maps can contain rich information, and the last concatenation further integrates different features. Through MB conv block, the number of channels increases but the size of feature maps declines; and moreover, the applied ACB block can avoid large amount of information loss during the down-sampling procedure. In following Eq. (2), how MB conv block works is described.

$$\begin{aligned} output_{mc} &= BP\left(AC_3^1\left(C_1\left(F_m\right)\right)\right) \oplus BP\left(AC_3^2\left(C_1\left(F_m\right)\right)\right) \\ &\oplus MP\left(C_1\left(F_m\right)\right), \quad where\ F_m = BP\left(C_3\left(F_i\right)\right) \end{aligned} \tag{2}$$

where $output_{mc}$ is the block output, F_i and F_m refer to input and intermediate feature maps, respectively; C_k $(k = 1, 3)$ represents $k \times k$ standard convolution, and AC_3^m indicates the ACB operation with kernel size of three and m repetition times; BP and MP stand for BN-PRelu and maximum pooling operations, respectively.

Following the MB conv block, a series of C-P blocks are placed to continuously mine deep semantic features of each modality, where the ACB and maximum pooling operations with skip connection are adopted, which can be expressed as follows:

$$\begin{aligned} output_{cp} &= F_m \oplus MP\left(BP\left(AC_3^1\left(F_m\right)\right)\right), \\ &where\ F_m = BP\left(AC_3^1\left(C_1\left(F_i\right)\right)\right) \end{aligned} \tag{3}$$

It is noticeable that each branch has been equipped with three C-P blocks, and output of each C-P block will serve as input of the multi-modal information fusion module (see Fig. 2).

At the end of the multi-scale feature extraction module, MB pool blocks are deployed to generate the final output in size of $1 \times 1 \times 256$ for all modalities. It should be pointed out that the designing of MB pool is derived from improvement on the multi-receptive field pooling block in [48]. To be specific, multi-branch down-sampling is used in MB pool to convert the small-size map into a feature vector, and original convolution used in [48] is replaced by the asymmetric one with small kernels, which can expand the depth and enhance the feature extraction. In following Eq. (4), work principle of the MB pool is depicted.

$$\begin{aligned} output_{mp} &= AP^2(BP(C_1(F_i))) \oplus AP^1(BP(AC_3^1(C_1(F_i)))) \\ &\oplus AP^1(C_1(F_i)) \end{aligned} \tag{4}$$

where $output_{mp}$ is the block output, and AP^i $(i = 1, 2)$ denotes that the average pooling operation is repeated for i times.

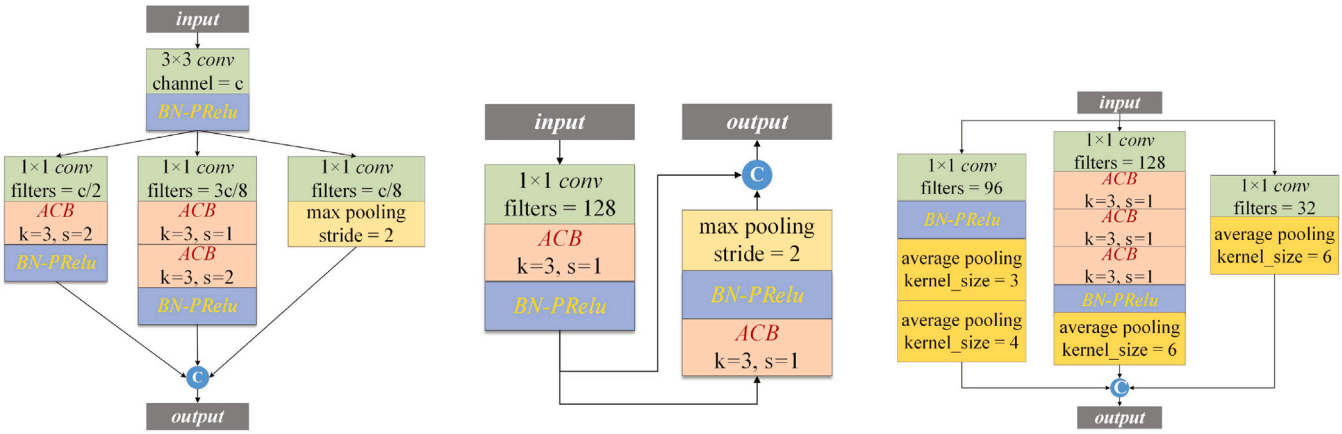
3.4. Multi-modal information fusion module

In the proposed AGGN, the multi-modal information fusion module is deployed to further integrate enhanced detailed and semantic features, and the structure is already illustrated in the green box of Fig. 2. In brief, fusion convolution realizes the integration of complementary advantages among the features of four modalities, where multi-scale feature maps in sizes of 24×24, 12×12 and 6×6 are fused. MB reduction block is adopted to transform the feature maps to vectors with strong presentation, which makes it feasible to further cascade the outputs of both multi-scale feature extraction and multi-modal information fusion module. MB reduction block consists of the sequential connection of the MB convolution (MC) and MB pool (MP) blocks, therefore, the block output $output_{mr}$ can be obtained by:

$$output_{mr} = MP(MC(F_i)) \tag{5}$$

where F_i denotes the input feature maps.

In addition, it is worth pointing out that the structure of fusion convolution block is similar to that of the C-P block, while the major difference is that the 2D convolution is replaced by the 3D one for fusion of feature maps with different modalities. According to Fig. 2, a finally constructed vector in size of $1 \times 1 \times 1792$ is fed into the last linear layers to obtain the glioma grading results, which is deemed to have strong presentation ability.



(a) MB conv (b) C-P (c) MB pool

Fig. 4. Architectures of blocks in multi-scale feature extraction module.

4. Results and discussions

In this section, the proposed AGGN is comprehensively evaluated on both internal and external public brain MRI datasets. In addition, substantial comparison experiments and ablation studies have been carried out to further validate the effectiveness and superiority of the proposed model. At first, experimental environment is briefly introduced.

4.1. Dataset preprocessing and experimental settings

The experimental data used in this paper come from the 2018 and 2019 brain tumor segmentation (BraTS) challenges organized by medical image computing and computer assisted intervention society (MICCAI) [49], which are collected by 3T MRI systems of 17 institutions. The dataset includes multi-modal MRI from 326 glioma patients (250 for HGG, 76 for LGG), in which each case contains 155 slice data of four modalities, and the original size of each image is 240 × 240. In addition, professional radiologists have annotated and calibrated the edema, necrosis and core areas of glioma to obtain tumor masks, and grading results are determined through further pathological analysis.

The dataset is divided into training set, testing set and validation set, where the training set and testing set are independent of each other, while the validation set is obtained by further splitting the training set. To be specific, ratio of the training and testing set is 2 : 1, where the images of training set come from 2018 BraTS. The testing set includes the internal and external subsets, which contain images from 2018 and 2019 BraTS, respectively. It should be pointed out that the main difference between internal and external testing subset is that samples of the latter belong to different data-source as those of training samples, and neither of the two subsets participates in the model training. Furthermore, one-fifth of the training samples are picked out to form the validation set for model tuning and selection.

In addition, preprocessing is performed on the initial data before training the model, where tumor masks are used to screen tumor-free slices at first, and it is noticeable that the selected slices without tumor are not fed into the subsequent process. For slices that contain tumor tissues, the foreground region is standardized and the proportion of background is reduced so that they are center-cropped to 192 × 192 in size; afterwards, four modalities are treated as four channels of the image. Finally, the dataset is divided according to the previously mentioned rules, and data augmentation operations are only performed on the training samples, including random rotation, translation and clipping. For a clear view, above preprocessing steps and dataset division are shown in Fig. 5.

All experiments in this study are carried out on the deep learning framework Pytorch, and the operating system is Windows 10 with NVIDIA GTX 2080Ti single GPU. Hyper-parameter settings are provided in Table 1, as for model parameters, initialization of convolution and fully-connected layers adopts the Kaiming method and normal distribution, respectively.

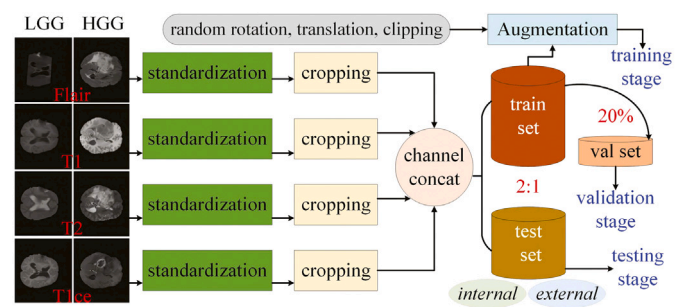


Fig. 5. Flowchart of preprocessing for brain MRI datasets.

Table 1
Hyperparameter settings.

Variables	Values
Training epochs	100
Batch size	32
Optimizer	Adam
Initial learning rate	0.0001
First-order moment decay coefficient	0.9
Second-order moment decay coefficient	0.999

4.2. Performance evaluation

To comprehensively evaluate performance of the proposed AGGN, four groups of experiments are carried out, which aim at verifying the generalization ability, architectural advantages, superiority against other representative CNN-based models and competitiveness in comparison to state-of-the-art glioma grading methods, respectively. Metrics *accuracy*, *precision*, *recall*, *specificity*, *F1 score* are adopted for performance evaluation, which can be calculated by following Eqs. (6)–(10):

$$accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{6}$$

$$precision = \frac{TP}{TP + FP} \tag{7}$$

$$recall = \frac{TP}{TP + FN} \tag{8}$$

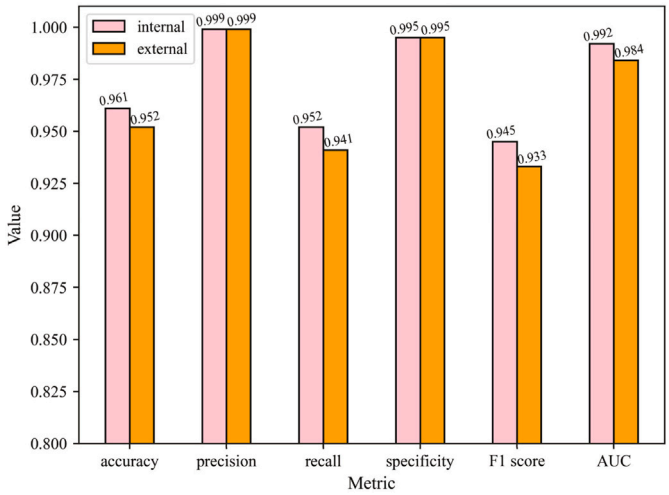


Fig. 6. Evaluation of AGGN on internal and external testing sets.

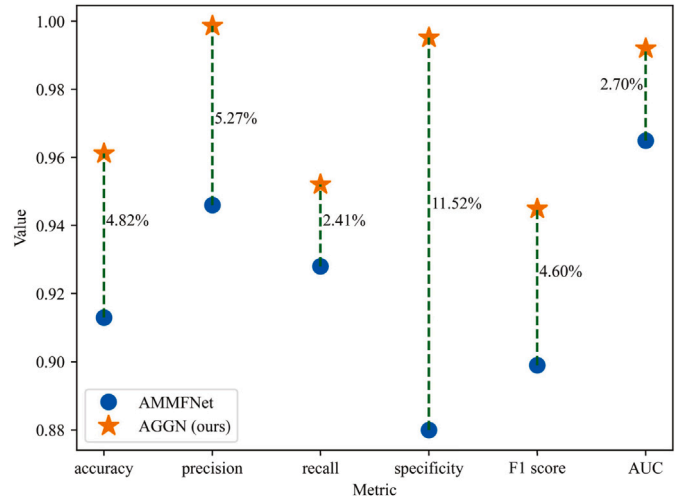


Fig. 7. Performance comparison between AGGN and AMMFNet.

$$specificity = \frac{TN}{FP + TN} \tag{9}$$

$$F1\ score = 2 \times \frac{recall \times precision}{recall + precision} \tag{10}$$

where TP/TN and FP/FN refer to the number of correct and wrong predictions of the HGG and LGG samples, respectively. As can be seen, the $accuracy$ describes the ratio of correct classifications of both HGG and LGG; the $precision$ aims at all samples predicted as HGG, and calculates the proportion of correct prediction; the $recall$ refers to the ratio of correctly identified HGG samples, which measures whether a model can screen all positive samples; similar to $recall$, the $specificity$ reflects the ability of identifying negative samples of a model; the $F1\ score$ takes the harmonic average between $accuracy$ and $recall$, and for all above five metrics, the larger their values are, the better the model performance is.

In addition, the receiver operating characteristic (ROC) curve and area under this curve (AUC) are also employed for the model evaluation. Specifically, the ROC curve takes the value of $1 - specificity$ (also known as false positive rate) as the horizontal axis and $recall$ as the vertical one, AUC is the area enclosed by ROC curve and the two coordinate axes.

Table 2
Performance comparison of proposed AGGN and four classic models on internal testing set.

Metrics	Models				
	[41]	[50]	[37]	[40]	AGGN
accuracy	0.9013	0.9038	0.8785	0.9330	**0.9612**
precision	0.7763	0.9632	0.8848	0.9687	**0.9987**
recall	0.9234	0.9119	0.9476	0.9181	**0.9521**
specificity	0.8320	0.8747	0.7235	0.9749	**0.9952**
F1 score	0.8687	0.8676	0.8506	0.9046	**0.9450**
AUC	0.9530	0.9570	0.9480	0.9780	**0.9920**

Table 3
Performance comparison of proposed AGGN and four classic models on external testing set.

Metrics	Models				
	[41]	[50]	[37]	[40]	AGGN
accuracy	0.8967	0.9052	0.8780	0.9446	**0.9519**
precision	0.9304	0.9672	0.8688	**0.9987**	**0.9987**
recall	0.9323	0.9122	**0.9658**	0.9325	0.9401
specificity	0.7890	0.8775	0.6937	0.9948	**0.9953**
F1 score	0.8614	0.8639	0.8503	0.9193	**0.9334**
AUC	0.9410	0.9570	0.9580	0.9700	**0.9840**

4.2.1. Generalization ability of AGGN

At first, results obtained by the proposed AGGN on both internal and external testing sets are shown in Fig. 6, and notice that in the former, training and testing samples share the same data-source; on the contrary, different sources are contained in the latter. As a result, this group of experiment can objectively reflect the generalization ability of the proposed AGGN. As is shown, the worst result is the $F1\ score$ on external dataset, which reaches 0.933; advantages of $precision$, $specificity$ and AUC are noticeable on both datasets, which validates that the propose AGGN is highly reliable in glioma grading task. At the same time, AGGN presents similar performance on internal and external testing sets, which demonstrates the robustness of AGGN in terms of handling various glioma MRI data; and moreover, this result indicates that AGGN can adapt to data from multi-center medical institution with strong generalization ability.

4.2.2. Architectural advantages of AGGN

In this part, to validate the architectural advantages of our method, the adaptive multi-modal fusion network (AMMFNet) [48] is adopted as baseline model for comparison, which is a similar glioma grading framework to the proposed AGGN. In Fig. 7, performance enhancement of AGGN on six indicators is illustrated, which shows that in comparison to AMMFNet, the proposed AGGN has improved all indicators to different extents. In particular, the most significant improvement is on $specificity$, which increases 11.52%. As previously mentioned, high $specificity$ is equivalent to low false positive rate. Consequently, it is verified that AGGN has strong ability to correctly identify negative samples, which can effectively avoid the waste of medical resources. In addition, $precision$ is increased by 5.27%, which implies that AGGN is able to achieve accurate diagnosis of HGG. In this group of experiment, the proposed AGGN presents noticeable competitiveness in comparison to similar framework, which indicates the advantages in structural configuration.

4.2.3. Comparisons with other CNN-based models

In order to further validate the competitiveness of the proposed AGGN, four other representative CNN-based models are adopted for comparison in this group of experiments, including ResNet-50 [41], DenseNet-101 [50], VGG-19 [37] and Inception-v4 [40]. For fairness, all models share the same training and testing data, and the results on internal and external datasets are reported in Tables 2 and 3, respectively. In addition, an illustration is presented in Fig. 8.

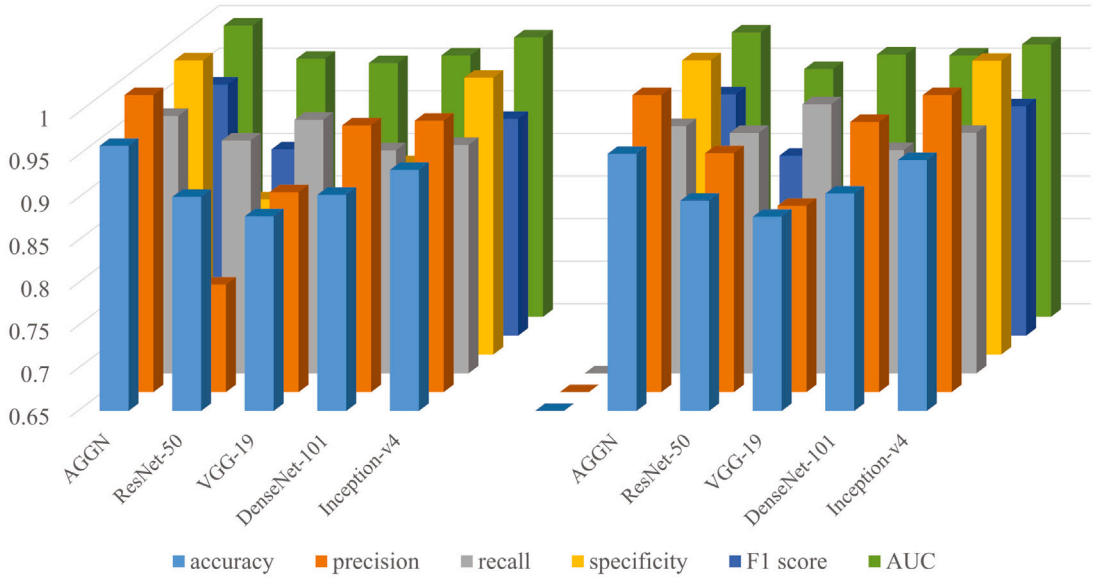


Fig. 8. Performance comparison of AGGN with advanced CNN-based models on internal (left) and external (right) testing sets.

As can be seen from Table 2, all the indicators of AGGN are better than those of other representative CNN models on internal dataset, which are 2.82%, 3.0%, 0.45%, 2.03%, 4.04% and 1.4% higher than the sub-optimal model on *accuracy*, *precision*, *recall*, *specificity*, *F1 score* and *AUC* respectively. While on the external testing set, the proposed AGGN also achieves satisfactory results of 95.19%, 99.87%, 94.01%, 99.53%, 93.34% and 98.40% on above six metrics, respectively. On five out of the six indicators, AGGN has obtained the best results.

In addition, the ROC curves with magnification on the two testing sets are presented in Fig. 9, which can effectively evaluate the diagnostic ability of a model and can maintain strong stability when the distribution of positive and negative samples changes. Notice that the curve close to the upper left corner has high prediction accuracy, and accordingly, the larger *AUC* value, the better model performance is. As shown in Fig. 9, the ROC curve of AGGN is above all other models on both the internal and external testing sets, and the *AUC* values reach 0.992 and 0.984, respectively. Consequently, the strong generalization ability of AGGN while facing different data sources is further verified.

Through this group of experiment, it is demonstrated that the proposed AGGN has overwhelming overall performance against other advanced CNN-based and domain-specific models on most metrics, which may owe to the meticulously designed and introduced dual-domain attention mechanism, multi-scale feature extraction and multi-modal information fusion modules.

4.2.4. Comparisons with state-of-the-art glioma grading algorithms

In this part, comparison between proposed AGGN and other state-of-the-art glioma grading algorithms are presented, including multistream CNN [51], multi-scale CNN [52], CAE-GAN (convolutional autoencoder and generative adversarial network) [36], pre-trained GoogleNet [33], 3DConvNet [32] and AMMFNet [48]. It should be pointed out that in previous experiments, data from the mentioned internal and external testing sets have none of the tumor masks, while most of recently related methods for the same task require the assistance of additional tumor masks. Consequently, to make a fair comparison, in this group of experiments, tumor masks have been added to original images for training, and the results are reported in Table 4. Notice that the data of other algorithms are cited from corresponding original papers, and "–" denotes none of relevant data is provided.

As can be found in Table 4, with assistance of tumor masks, the proposed AGGN can present the state-of-the-art performance. In particular, the *AUC* value of AGGN with and without tumor masks are 0.998 and 0.992, respectively, which implies that the assistance of tumor masks does further improve the model performance. It is also worth mentioning that the *AUC* of 0.992 (without masks) is already an excellent result. Hence, it can be concluded that performance of AGGN has little reliance on the masks, which demonstrates that AGGN can overcome the high dependence of manually labeled annotations so as to achieve the end-to-end applications in practice.

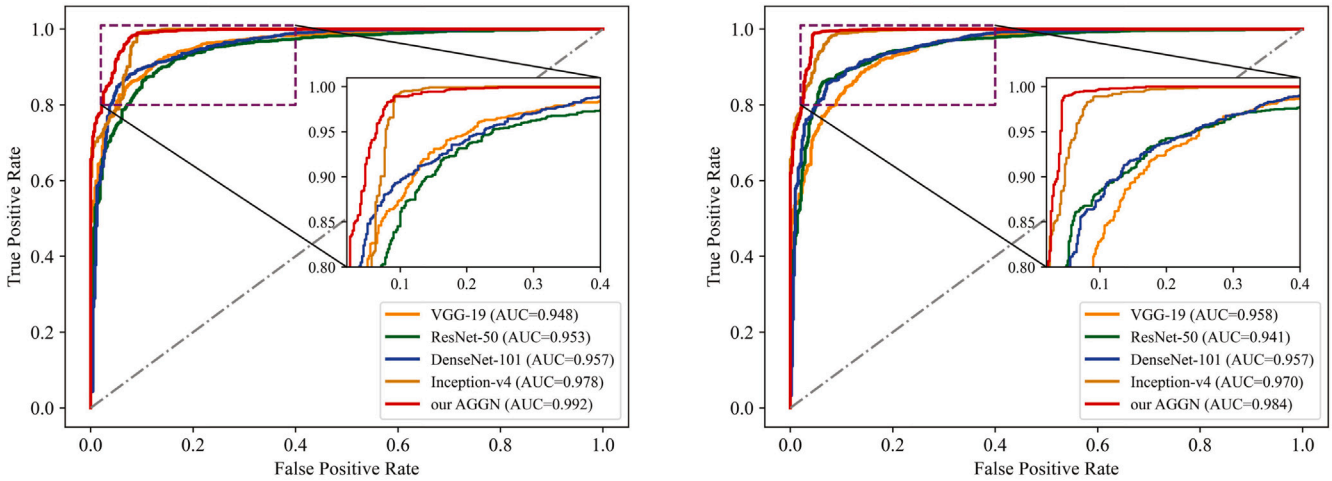
Table 4
Performance of AGGN and other advanced models on tumor mask assisted data.

Models	Metrics			
	accuracy	*recall*	*specificity*	*AUC*
Multistream CNN	0.9087	–	–	–
Multi-scale CNN	0.8947	–	–	–
CAE-GAN	0.9204	–	–	–
Pre-trained GoogleNet	0.9450	–	–	0.9680
3DConvNet	0.9710	0.9470	0.9680	–
AMMFNet	0.9820	**1.0000**	0.9330	0.9970
AGGN (ours)	**0.9899**	**1.0000**	**0.9678**	**0.9980**

4.3. Ablation study

To validate the effectiveness of core components in the proposed AGGN, substantial ablation studies are performed on the internal testing set in this subsection. The designed dual-domain attention mechanism is firstly verified and the results are reported in Table 5, where AGG1, AGG2 and AGG3 refer to the model with none of attention modules, only spatial and only channel attention module, respectively. Obviously, in comparison to AGG1, on most indicators the performance has been improved to a certain extent after introducing the spatial or channel attention mechanism. It is also found that the applied dual-domain attention module in the proposed AGGN has realized significant performance enhancement, which improves *accuracy*, *recall*, *F1 score* and *AUC* by 2.92%, 3.61%, 4.23% and 1.6%, respectively.

In addition, the ROC curves of four models listed in Table 5 are presented in Fig. 10, where it can be seen that the proposed AGGN has obtained the best results. In particular, when false positive rate is 0, the true positive rate of AGGN is close to 0.85 and *AUC* is 0.992, which implies that the proposed AGGN can accurately identify HGG with almost none of false detection. Therefore, the proposed AGGN is a reliable model that can provide a solid guarantee for the diagnosis and treatment of critical patients.



(a) On internal testing set (b) On external testing set

Fig. 9. ROC curves of the proposed AGGN and other CNN models.

Table 5
Ablation studies of dual-domain attention mechanism on internal testing set.

Models	Metrics					
	accuracy	*precision*	*recall*	*specificity*	*F1 score*	*AUC*
AGG1	0.9320	1.0000	0.9160	1.0000	0.9027	0.9760
AGG2	0.9481	0.9866	0.9467	0.9532	0.9270	0.9860
AGG3	0.9476	**1.0000**	0.9349	**1.0000**	0.9238	0.9900
AGGN	**0.9612**	0.9987	**0.9521**	0.9952	**0.9450**	**0.9920**

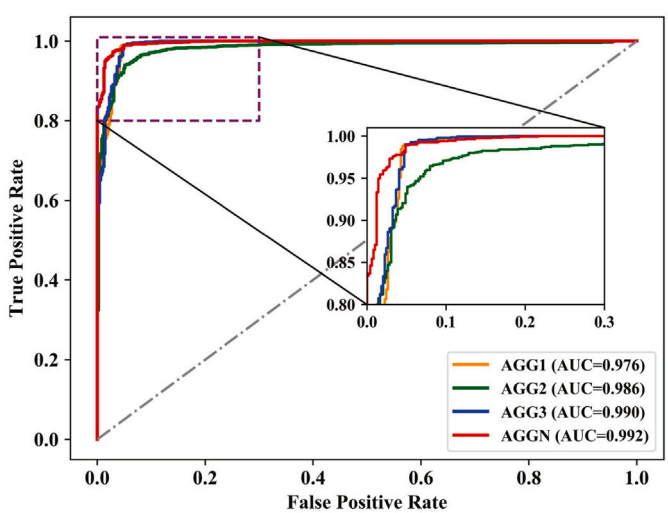


Fig. 10. ROC curves of AGG1, AGG2, AGG3, and AGGN.

According to above results, effectiveness of the dual-domain attention mechanism is sufficiently validated. Before extracting multi-scale features, channel-domain attention is firstly introduced to low-level detail information, which determines what deserves attention in each modality of brain MRI; afterwards, spatial-domain attention is used to learn spatial dependence among high-level semantic information, so as to figure out the important locations in feature maps. As a result, the proposed AGGN can both recognize and localize the significant pathological features in brain MRI with strong robustness.

In the following, ablation study results on multi-branch convolution block and the multi-modal information fusion module are reported in Table 6. It should be pointed out that except for the investigated components, other configurations of AGGN remain unchanged so as to make objective and convincing comparisons.

Firstly, as the most important component of the multi-scale feature extraction module, MB conv block is compared with multi-receptive field (MRF) conv block of AMMFNet. As can be seen from Table 6, the MB conv block has overwhelmed the MRF conv block on all metrics, which demonstrates that the multi-branch structure has superiority in dealing with glioma grading task based on brain MRI. Further explorations of the essential mechanism show that the volume of glioma from patients can be quite different, whereas the proposed MB conv block has adopted ACBs with different sizes to extract image features in parallel, which can obtain fine-grained texture and tissue information of multi-modal brain MRI. Therefore, it can be inferred that the multi-scale feature extraction module has made great contributions to the overall model performance.

Secondly, the designed multi-modal information fusion module is compared with the approach in [48]. As reported in Table 6, on four out of six indicators, proposed AGGN has achieved slight performance improvement. Specifically, AGGN improves $accuracy$, $recall$, $F1\ score$ and AUC by 3.4%, 0.6%, 0.3% and 0.7%, respectively. It is worth mentioning that in the proposed AGGN, final input vector to the classifier is a concatenation of outputs from seven branches, and this number is fewer than that in [48]. Consequently, it can be concluded that AGGN has achieved comparable results to the model in [48] with a simplified structure. Additionally, the concatenation manner in AGGN has avoided stacking redundant features, which not only benefits highly-efficient feature fusion, but also simultaneously avoids excessively complicating the model.

4.4. Computational complexity analysis

In this study, the number of model parameters (Params) and floating point operations (FLOPs) are adopted to depict the spatial and time complexity of the proposed AGGN, respectively. Excessive parameters will impede the light-weight deployment of model on edge devices, and too large FLOPs will influence the convergence during model training, which directly determines the accuracy of the model inference.

On the one hand, Params of the proposed AGGN is 16.37M, which is 9.13M fewer than that of the classical ResNet-50 model. According to Table 3, the accuracy of AGGN is even 5.52% higher than that of ResNet-50, which demonstrates that the developed AGGN can effectively balance the computational costs and accuracy. It may owe to the proposed AGGN has effectively reduced the parameters by replacing large-sized kernels with a series of small-sized ones. Meanwhile, the

Table 6
Ablation studies of MB conv block and information fusion methods.

Models	Metrics					
	accuracy	*precision*	*recall*	*specificity*	*F1 score*	*AUC*
MRF conv block of AMMFNet	0.9577	0.9960	0.9502	0.9856	0.9400	0.9840
MB conv block of AGGN	**0.9612**	**0.9987**	**0.9521**	**0.9952**	**0.9450**	**0.9920**
Fusion method in [48]	0.9578	**1.0000**	0.9461	**1.0000**	0.9420	0.9850
Fusion method of AGGN	**0.9612**	0.9987	**0.9521**	0.9952	**0.9450**	**0.9920**

accuracy of AGGN is mainly guaranteed by the structural advantages, including employing dual-domain attention mechanism to highlight key features, realizing feature extraction on each individual modality, and integrating multi-modal information in different levels.

On the other hand, the FLOPs of AGGN are 24,790M, which mainly due to the large size of multi-channel input samples, where the data processing has consumed great deals of the FLOPs. It is also worth mentioning that during the model training, none of obvious overfitting phenomenon has occurred, which implies that the training and inference time consumed by the proposed AGGN is acceptable.

To sum up, the proposed AGGN can effectively achieve the balance between model complexity and accuracy, which has achieved satisfactory results in the glioma grading task with considerable efficiency.

5. Conclusion

In this paper, a novel self-attention based network AGGN has been developed, which mainly consists of three meticulously designed modules, including a dual-domain attention module, a multi-scale feature extraction and a multi-modal information fusion one. Robust features with strong presentation ability are constructed by integrating outputs of the latter two modules, which are used to eventually realize the glioma grading task. Performance of the proposed AGGN has been comprehensively evaluated on both internal and external testing sets, and the results have demonstrated the superiority of AGGN against other state-of-the-art algorithms. Furthermore, substantial ablation studies have verified effectiveness of the designed three modules in AGGN, which can take full advantages of detailed and semantic information so that model performance can be greatly improved, and simultaneously computational burdens are released to some extent.

Although the proposed AGGN has presented satisfactory performance on the glioma grading task, it still has some spaces for further improvement, including task migration adaption, quantitative lesion analysis, and model lightweighting studies. In future work, we aim to (1) apply the developed AGGN framework to other MRI-based tasks such as stroke and cancer diagnosis; (2) investigate fine-grained glioma grading methods to support quantitative analysis; (3) further optimize the structure of AGGN through fuzzy system and tensor decomposition techniques. [53–56]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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